LISBON TEMPERATURE FORECASTING: Applying the Box-Jenkins Methodology

Duarte Carvalho (20221900), Matilde Casimiro (20221940), Laura Matias (20221836)

NOVA IMS Information Management School

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Introduction

Lisbon, the capital of Portugal, experiences a diverse array of climatic changes throughout the year, significantly impacting its urban lifestyle and economic sectors. This study investigates the temperature trends in Lisbon from January 1995 to May 2020, employing the **Box-Jenkins Methodology** — a robust approach for time series analysis and forecasting. Through meticulous analysis, the aim is to uncover seasonal patterns and accurately predict future climatic tendencies.

After transforming the data into a tsibble object, an initial graph was created to visualize temperature patterns and trends. The analysis reveals a clear seasonal pattern, with temperatures rising during the warm summers and dropping in the cool winters, ranging from about 5°C to 25°C. Interestingly, the long-term data does not show a significant trend towards getting warmer or cooler, indicating a stable climate over this period.

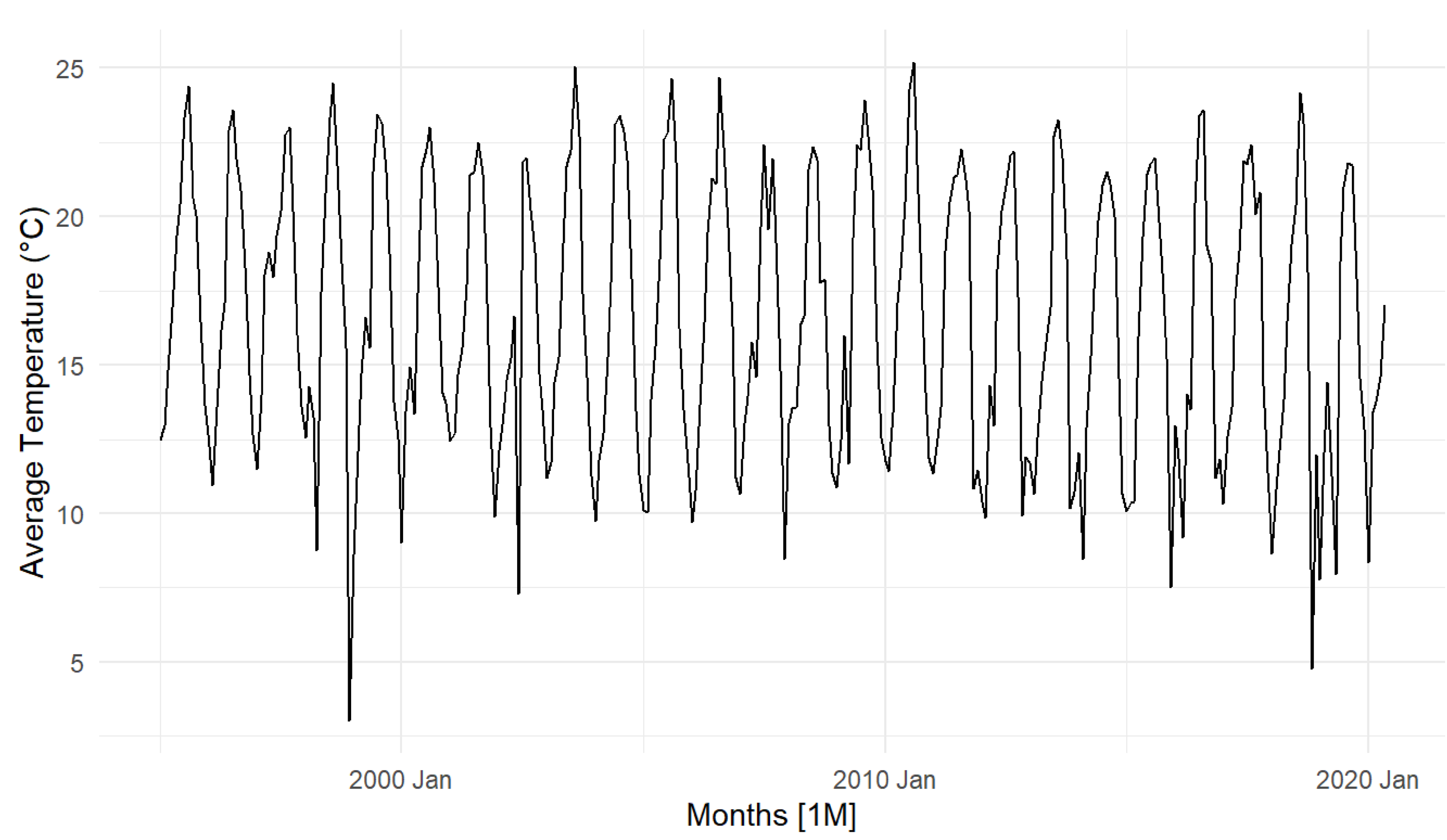

Figure 1: Average Temperature by Month in Lisbon

Methodology

For validation purposes, the data was partitioned into two distinct subsets: the training set, spanning from January 1995 to December 2017, and the test set, encompassing the period from January 2018 to May 2020. Prior to partitioning, the **Augmented Dickey-Fuller (ADF) Test** was conducted to evaluate the stationarity of the time series. The results confirmed the non-stationary nature of the data. To address this issue, seasonal differencing was employed, which effectively made the series stationary.

Following this, the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were analyzed to identify potential candidate models. The ACF exhibited prominent spikes at lag 1 and seasonal lag 12, indicating potential seasonal and non-seasonal moving average components. Simultaneously, the PACF displayed significant spikes at lag 1 and seasonal lags 12, 24, and 36, suggesting potential seasonal and non-seasonal autoregressive components.

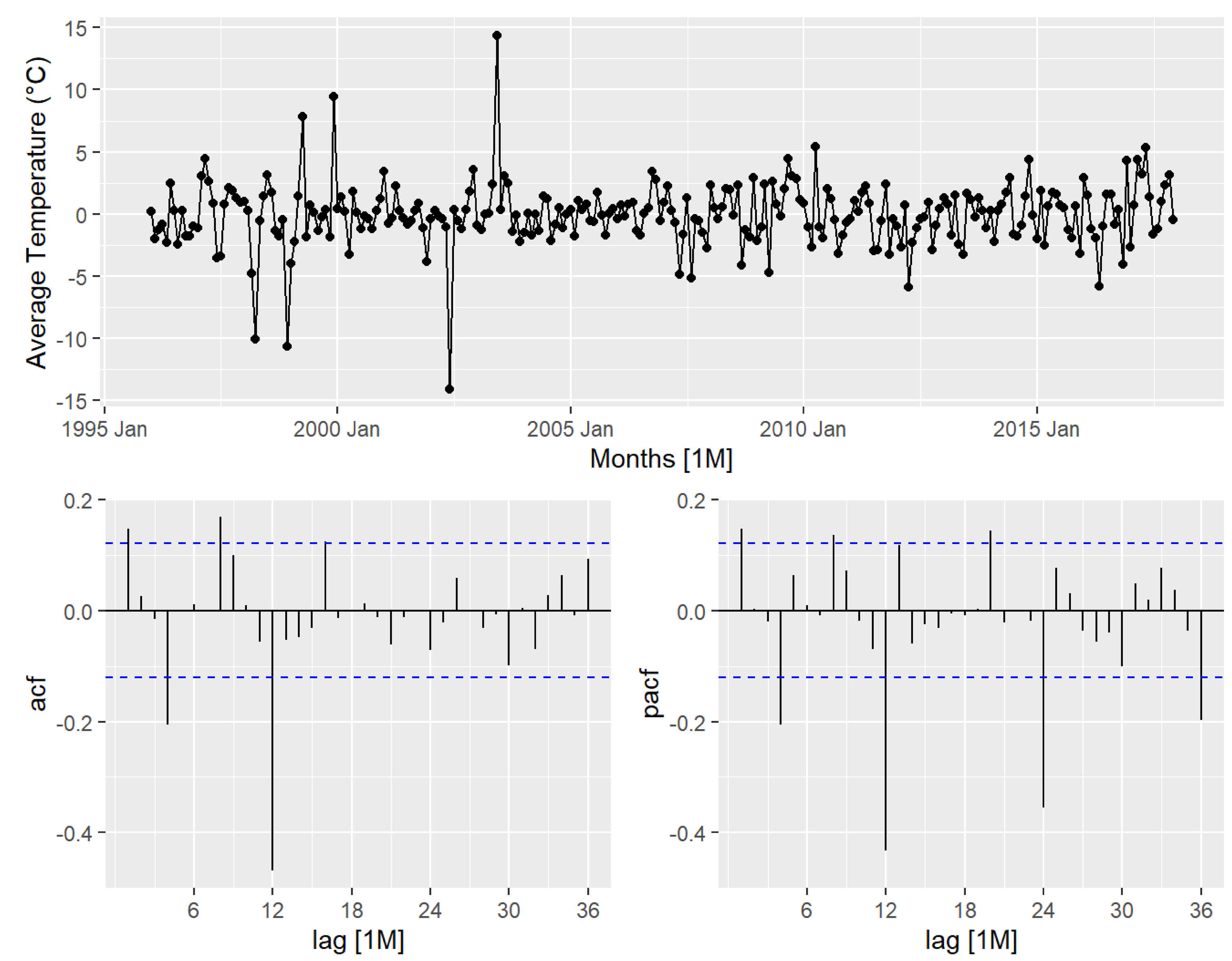

Figure 2: ACF and PACF of the Seasonal Difference

Results

To identify the optimal model, a thorough evaluation of various models was conducted, focusing on selecting the five models with the lowest AIC and BIC values for further analysis: SARIMA(1,0,0)(0,1,1), SARIMA(0,0,1)(0,1,1), SARIMA(1,0,0)(1,1,1), SARIMA(1,0,1)(1,1,1), and SARIMA(0,0,2)(0,1,1). These selections were made based on their ability to minimize information criteria. Subsequently, the residuals of these chosen models underwent the **Ljung-Box Test**. The results indicated that none of the models exhibited significant autocorrelation in their residuals, suggesting that these residuals possess properties consistent with white noise. This implies that the selected models effectively capture the underlying patterns within the data.

Following the evaluation, forecasts generated for the top five models were presented in Table 1. Among these, the **SARIMA(1,0,0)(1,1,1)** model was particularly noteworthy. A closer examination of the data confirmed its standout performance, with the lowest RMSE indicating a strong fit to the data. While its MAE is not the lowest, it remains relatively low, which is a positive indicator. The MPE is closer to zero compared to other models, suggesting a more accurate prediction without bias. Although the MAPE is slightly higher, it is still within an acceptable range.

Table 1: Error Measures on the Test Set

.model	.type	RMSE	MAE	MPE	MAPE
sarima001_011	Test	2.938601	1.959119	-16.71329	21.01755
sarima002_011	Test	2.938601	1.959051	-16.71397	21.01721
sarima100_011	Test	2.938958	1.960688	-16.70721	21.02819
sarima100_111	Test	2.931141	1.968256	-16.41727	21.03683
sarima101_111	Test	2.933392	1.968848	-16.44137	21.05304

Conclusion

In light of the comprehensive analysis, the *SARIMA(1,0,0)(1,1,1)* model emerged as the superior choice, striking an optimal balance between adherence to the training data and predictive precision for the test set. This model was selected for its adeptness in capturing the intricate seasonal patterns observed in the data. The forecasts generated by this model are depicted in Figure 3, showcasing its prediction prowess and the anticipated climatic trends for Lisbon. But here's the thing: our climate is changing. Because of global warming, temperatures are all over the place, and what used to look like regular patterns might not be so reliable anymore. This change is making it more difficult for us to forecast the future, and despite the strength of the model, unforeseen deviations or anomalies in temperature trends may occur due to the ongoing changes in our climate.

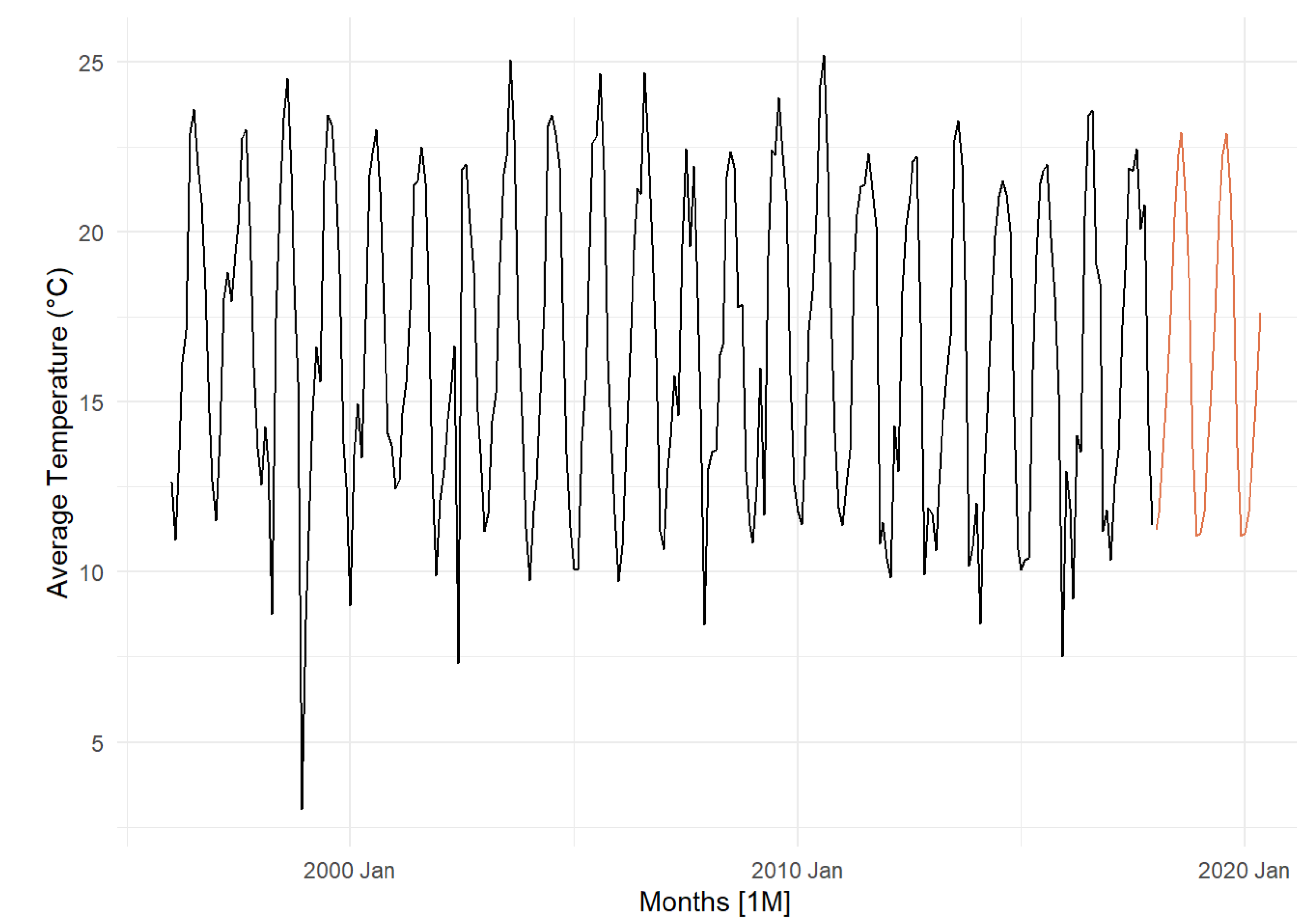

Figure 3: Forecasted Average Temperature in Lisbon

References

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[2] GeeksforGeeks. "Box-Jenkins Methodology for ARIMA Models." GeeksforGeeks, 5 Apr. 2020, https://www.geeksforgeeks.org/box-jenkins-methodology-for-arima-models/.

[3] Kaggle. "Temperature of All Countries (1995-2020)." Kaggle, Subham Jain, https://www.kaggle.com/datasets/subhamjain/temperature-of-all-countries-19952020.